



Consideraciones día 2

**Taller de secuenciación
genómica y análisis
bioinformático para
patógenos bacterianos
con enfoque una salud.**

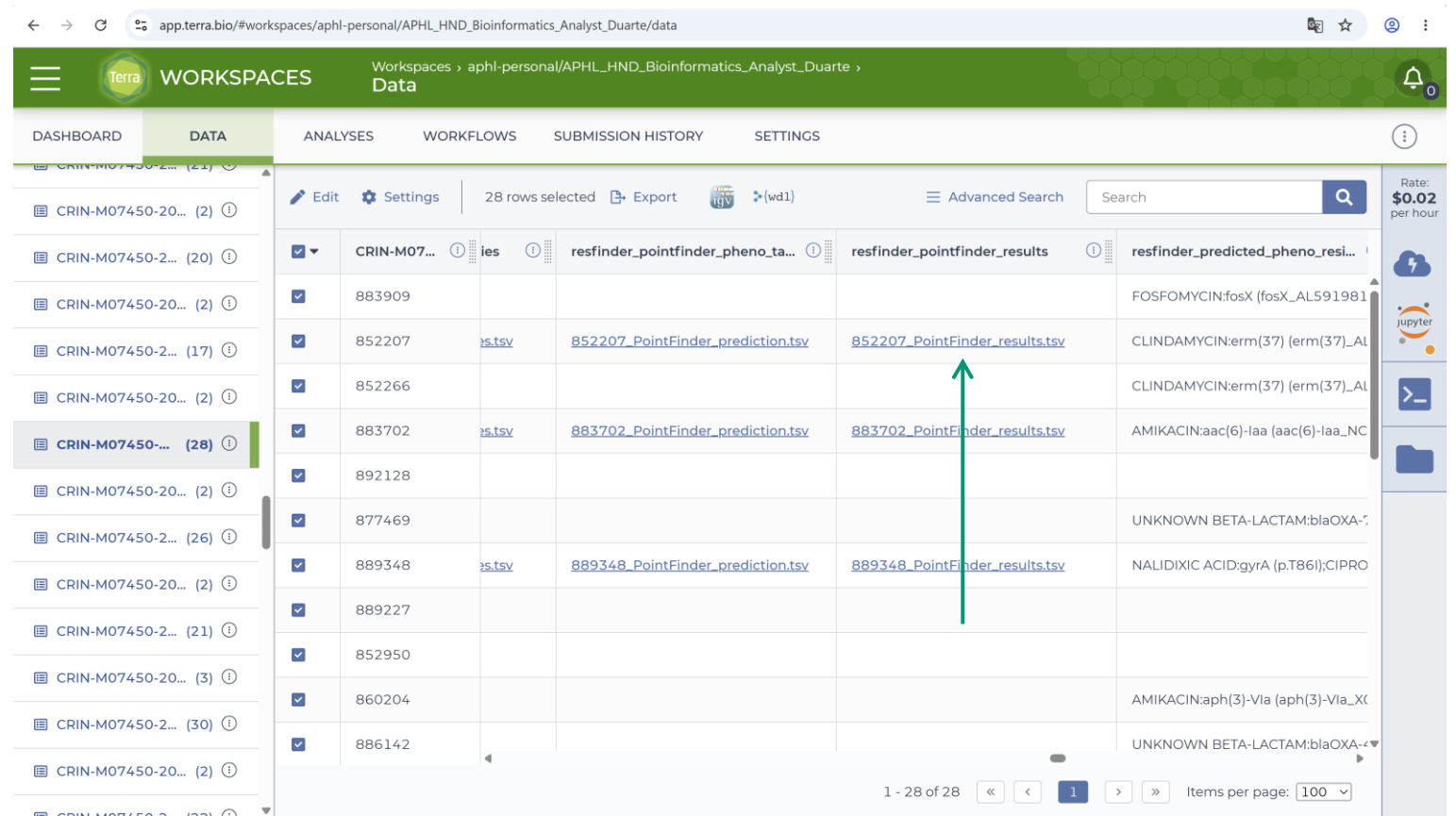
**Laboratorio de genómica y
biología molecular,
CNRIMA. Inciensa**

Junio, 2026

Resultados Terra

Pointfinder

- No aparece en la tabla resumen
- Se debe descargar de la aplicación el archivo con las mutaciones puntuales



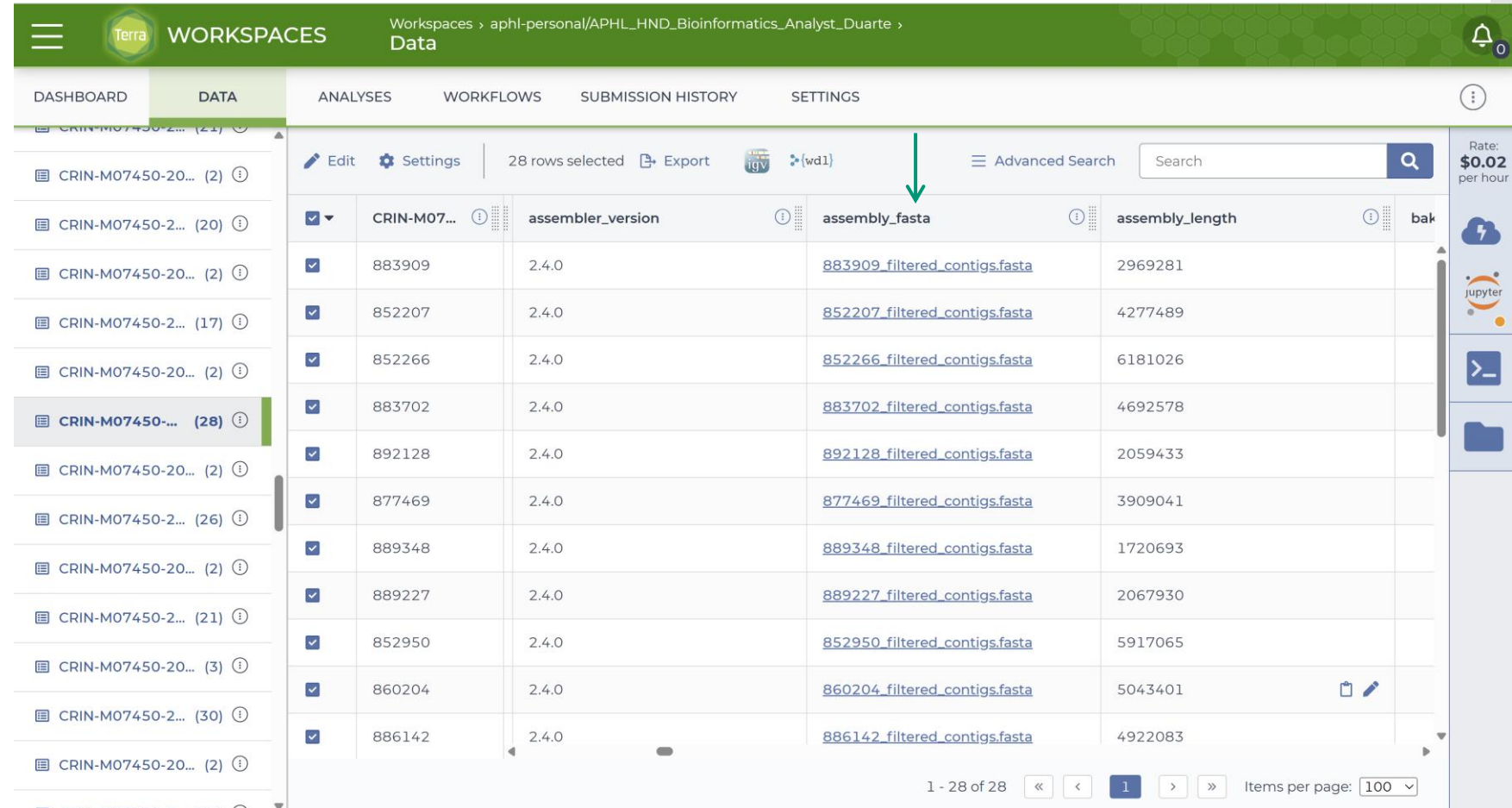
The screenshot shows the Terra web interface with a table of Pointfinder results. The table has five columns: 'CRIN-M07...', 'resfinder_pointfinder_pheno_ta...', 'resfinder_pointfinder_results', and 'resfinder_predicted_pheno_resi...'. A green arrow points to the 'resfinder_pointfinder_results' column. The table contains 28 rows of data, with the first row highlighted. The interface includes a sidebar with a list of datasets, a top navigation bar, and a right sidebar with a search bar and a 'Rate: \$0.02 per hour' indicator.

CRIN-M07...	resfinder_pointfinder_pheno_ta...	resfinder_pointfinder_results	resfinder_predicted_pheno_resi...
883909			FOSFOMYCIN:fosX (fosX_AL591981
852207	852207_PointFinder_prediction.tsv	852207_PointFinder_results.tsv	CLINDAMYCIN:erm(37) (erm(37)_AL
852266			CLINDAMYCIN:erm(37) (erm(37)_AL
883702	883702_PointFinder_prediction.tsv	883702_PointFinder_results.tsv	AMIKACIN:aac(6)-Iaa (aac(6)-Iaa_NC
892128			
877469			UNKNOWN BETA-LACTAM:blaOXA-7
889348	889348_PointFinder_prediction.tsv	889348_PointFinder_results.tsv	NALIDIXIC ACID:gyrA (p.T86I);CIPRO
889227			
852950			
860204			AMIKACIN:aph(3)-Via (aph(3)-Via_XI
886142			UNKNOWN BETA-LACTAM:blaOXA-4

Resultados Terra

Otros descargables

- Fastq trimmed: **read1_clean**, **read2_clean**
- **Ensamblaje: (Genoma borrador) Archivo fasta:**



The screenshot shows the Terra Workspaces interface. The top navigation bar includes 'WORKSPACES' and a breadcrumb trail: 'Workspaces > aphl-personal/APHL_HND_Bioinformatics_Analyst_Duarte > Data'. Below this is a tabbed interface with 'DATA' selected. The main content area displays a table with 28 rows selected. A green arrow points to the 'assembly_fasta' column. The table has columns for selection, workspace ID, assembler version, assembly FASTA file, assembly length, and a backup column. The FASTA files are named with the workspace ID followed by '_filtered_contigs.fasta'.

		assembler_version	assembly_fasta	assembly_length	bak
☒	CRIN-M07...				
☒	883909	2.4.0	883909_filtered_contigs.fasta	2969281	
☒	852207	2.4.0	852207_filtered_contigs.fasta	4277489	
☒	852266	2.4.0	852266_filtered_contigs.fasta	6181026	
☒	883702	2.4.0	883702_filtered_contigs.fasta	4692578	
☒	892128	2.4.0	892128_filtered_contigs.fasta	2059433	
☒	877469	2.4.0	877469_filtered_contigs.fasta	3909041	
☒	889348	2.4.0	889348_filtered_contigs.fasta	1720693	
☒	889227	2.4.0	889227_filtered_contigs.fasta	2067930	
☒	852950	2.4.0	852950_filtered_contigs.fasta	5917065	
☒	860204	2.4.0	860204_filtered_contigs.fasta	5043401	 
☒	886142	2.4.0	886142_filtered_contigs.fasta	4922083	

Y mucho más.....

Con los
ensamblajes
(salida de Terra
o de Unicycler)
podemos:

Herramientas Web:

- **PubMLST:** Identificación por rMLST y subtipificación por MLST

<https://pubmlst.org/>

- **Center for genomic epidemiology_ CGE:** SeqSero, Resfinder, Virulencefinder, pointfinder...y más....

<https://www.genomicepidemiology.org/>

- **Skani:** Cálculo de ANI:

<https://gtadb.ecogenomic.org/tools/skani>

- **Pathogenwatch:** Identificación y caracterización de microorganismos

<https://pathogen.watch/en>

Y mucho más.....

Steps to Build a Phylogeny (Very Simplified)

FASTQ

FASTA

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2. Define the dataset: Select appropriate sequences based on the question.

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4. Align sequences: you could use MAFFT; verify and correct alignment (Aliview).

5. Reconstruct the phylogenetic tree: Recommended software: IQ-TREE.

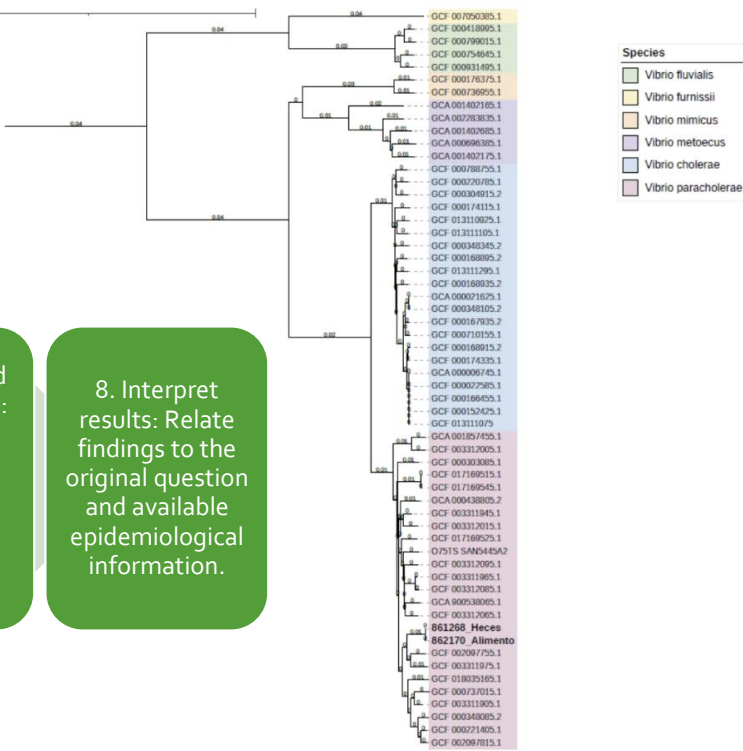
6. Evaluate tree reliability: Perform bootstrap or other support analyses; interpret node support values.

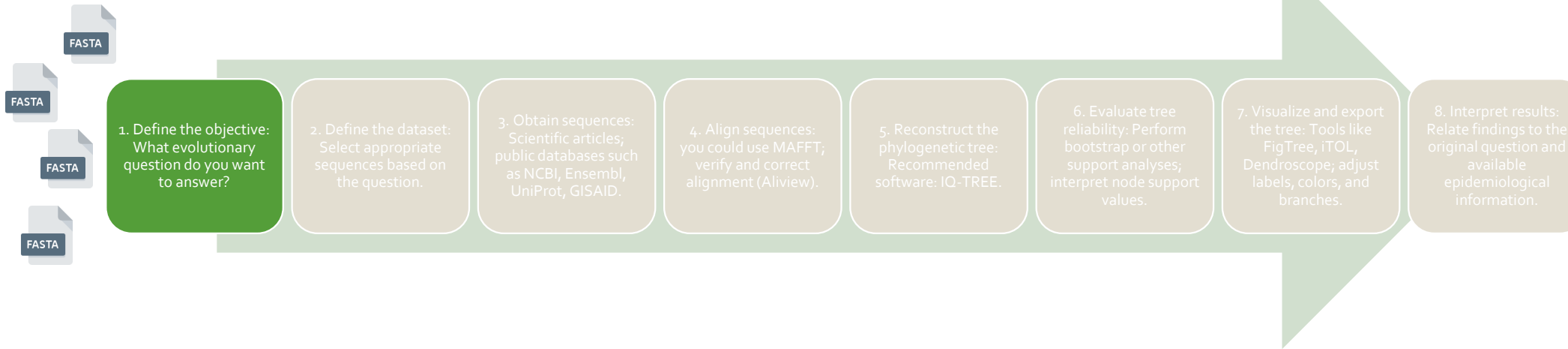
7. Visualize and export the tree: Tools like FigTree, iTOL, Dendroscope; adjust labels, colors, and branches.

8. Interpret results: Relate findings to the original question and available epidemiological information.

9. Communicate the results.

Tree scale: 0.1



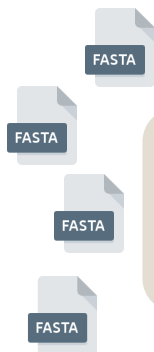


Tipo de actividad:
Secuenciación de genoma completo y análisis bioinformático de las muestras Inciens # 861268 y # 862170 correspondientes a cepas bacterianas identificadas como *V. cholerae*, aisladas a partir de una muestra clínica de heces de paciente y de una muestra de licuado de verduras respectivamente, ambas muestras estaban asociadas a un evento de enfermedad diarreica aguda ocurrido en el centro de cuido Fundación [redacted] a finales de abril del año en 2025.

Cuadro 1. Identificación a nivel de especie de las secuencias obtenidas por cuatro herramientas bioinformáticas diferentes.

Muestra	Herramienta bioinformática			
	ani_top_species_match	gambit_predicted_taxon	kmerfinder_top_hit	Ribosomal MLST
861268	<i>Vibrio_cholerae</i>	<i>Vibrio paracholerae</i>	<i>Vibrio cholerae</i>	<i>Vibrio paracholerae</i>
862170	<i>Vibrio_cholerae</i>	<i>Vibrio paracholerae</i>	<i>Vibrio cholerae</i>	<i>Vibrio paracholerae</i>

- ¿La muestra del paciente y de [redacted] alimento están genéticamente relacionadas entre sí?.
- ¿Cuál es la identificación a nivel de especie del aislamiento?



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Population Analysis of *Vibrio cholerae* in Aquatic Reservoirs Reveals a Novel Sister Species (*Vibrio paracholerae* sp. nov.) with a History of Association with Humans

Mohammad Tarekul Islam ^a, Tania Nasreen ^a, Paul C. Kirchberger ^b, Kevin Y. H. Liang ^a,
Fabini D. Orata ^a, Fatema-Tuz Johura ^c, Nora A. S. Hussain ^a, Monica S. Im ^d, Cheryl L. Tarr ^d,
Munirul Alam ^c, Yann F. Boucher ^{a,e}

^aDepartment of Biological Sciences, University of Alberta, Edmonton, Alberta, Canada

^bDepartment of Integrative Biology, University of Texas at Austin, Austin, Texas, USA

^cInfectious Diseases Division, International Centre for Diarrheal Disease Research, Bangladesh (icddr), Dhaka, Bangladesh

^dNational Center for Emerging and Zoonotic Infectious Diseases, Centers for Disease Control and Prevention, Atlanta, Georgia, USA

^eSaw Swee Hock School of Public Health, National University of Singapore, Singapore

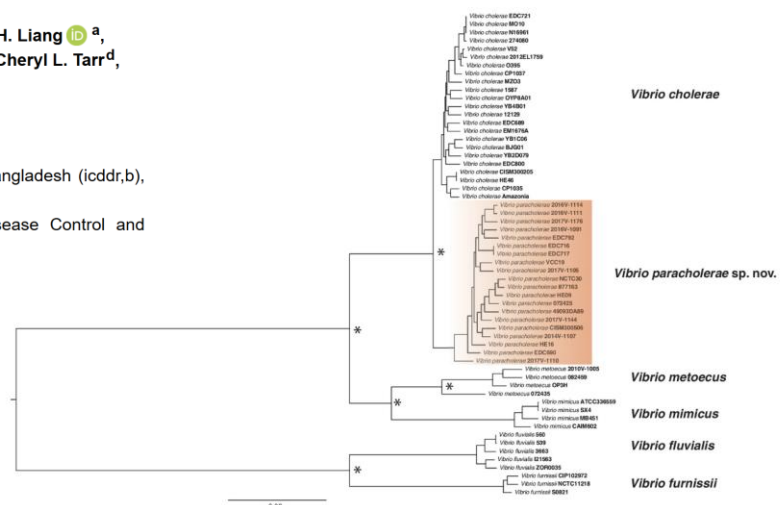
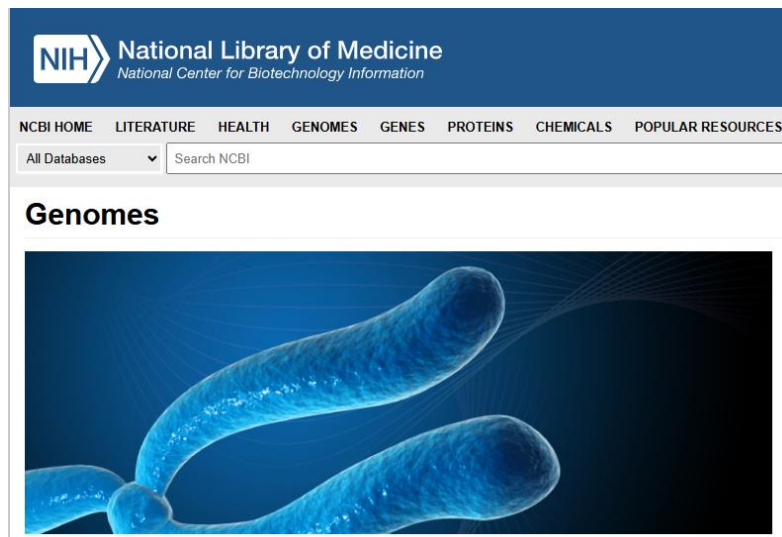
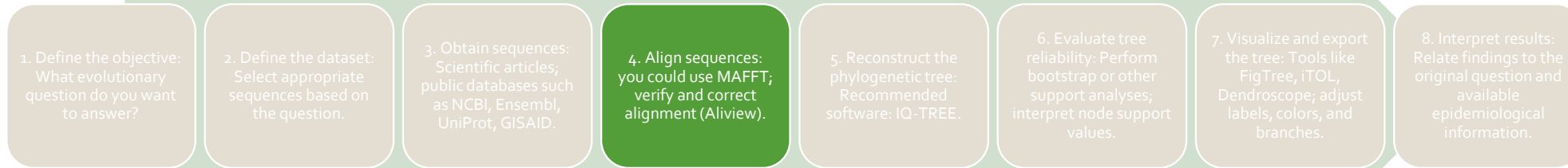
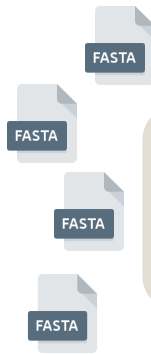
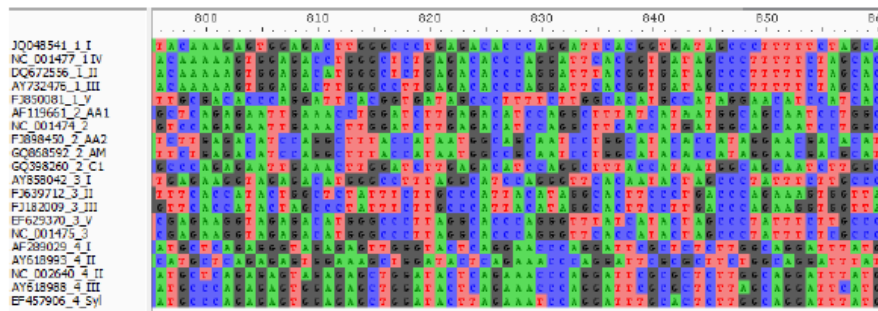


FIG 5 Whole-genome phylogenetic tree of *Vibrio paracholerae* along with its closest relatives. The maximum likelihood phylogenetic tree was constructed from the core genome alignment of ≈ 2.1 Mbp using the GTR gamma substitution model. Corresponding nodes with relevant bootstrap support over 70% from the 100 replicates were indicated with an asterisk (*). The scale bar represents nucleotide substitutions per site.

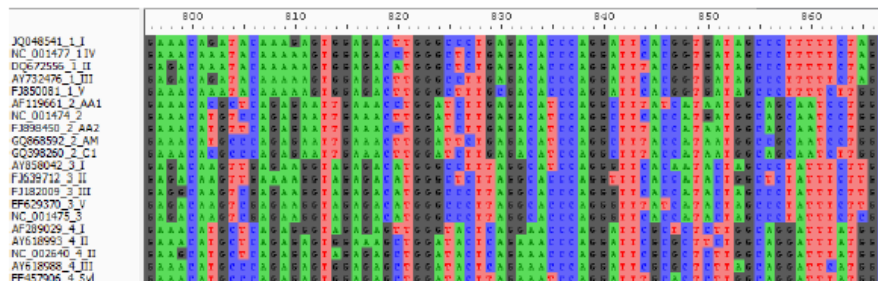




✓ Positional homology

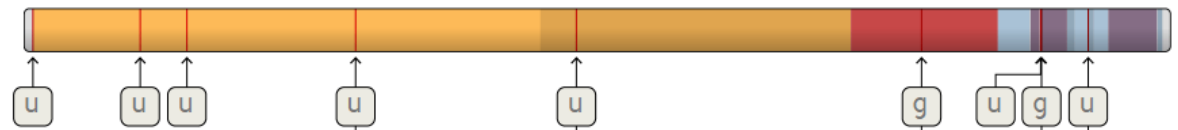


Non-aligned sequences



Aligned sequences

- Los perfiles de SNPs son “huellas digitales” genéticas
- SNP = Single Nucleotide Polymorphism
 - Secuencia: ATGTTCTC
 - Referencia: ATGTTGCTC
- Los SNP ocurren en todo el genoma, con frecuencia variada:



- Para poder compararlas se deben combinar los perfiles de SNPs en una alineación de secuencia múltiple (MSA) de múltiples genomas
- Los MSA se utilizan para:
 - Medir las relaciones entre secuencias
 - Construir árboles filogenéticos

Alineamiento de secuencias múltiples

```
AGGTACTAAGCC
AGGTACTATGCC
AGCTACTATGCC
AGGTACTATGCG
AGCTACTATGGC
TGCTACTATGCC
-AGCTACTATGGC
AGCTACTATGGC
- -TGCTACTATGCC
AGGAACTATGCG
AGGAACTATGCG
```



ClustalW/X:
Progressive
alignment.

Muscle: Iterative
method.



MAFFT: multiple
methods.

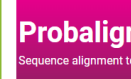
T-Coffee



PROBCONS

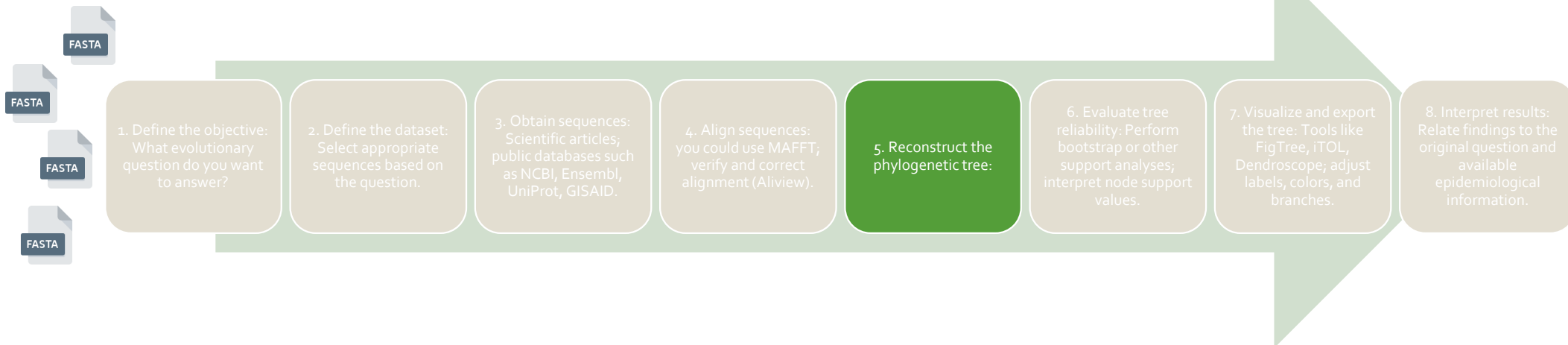
ProbCons

Probalign



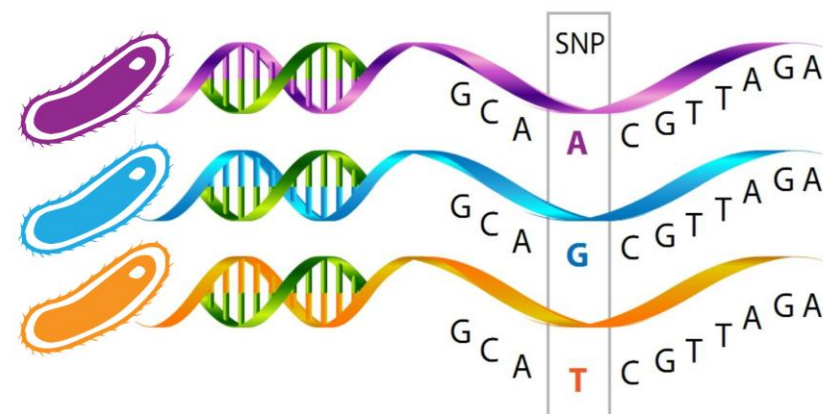
```
AGGTACTAAGCC
AGGTACTATGCC
AGCTACTATGCC
AGGTACTATGCC
AGCTACTATGCC
TGCTACTATGCC
AGCTACTATGCC
AGCTACTATGCC
TGCTACTATGCC
AGGAACTATGCC
AGGAACTATGCC
```



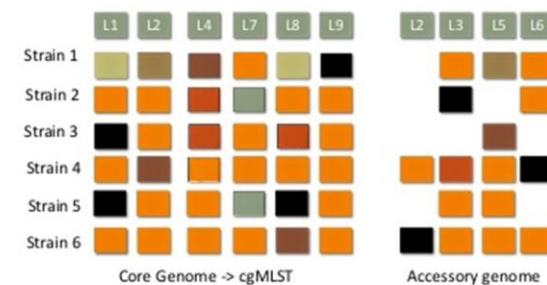


1. Identificación de diferencias genéticas

- **Polimorfismo de un solo nucleótido (SNP)**
 - Identificación de variaciones nucleotídicas
- **Core/whole-genome Multilocus sequence types (cg/wgMLST)**
 - Identificación de variaciones alélicas

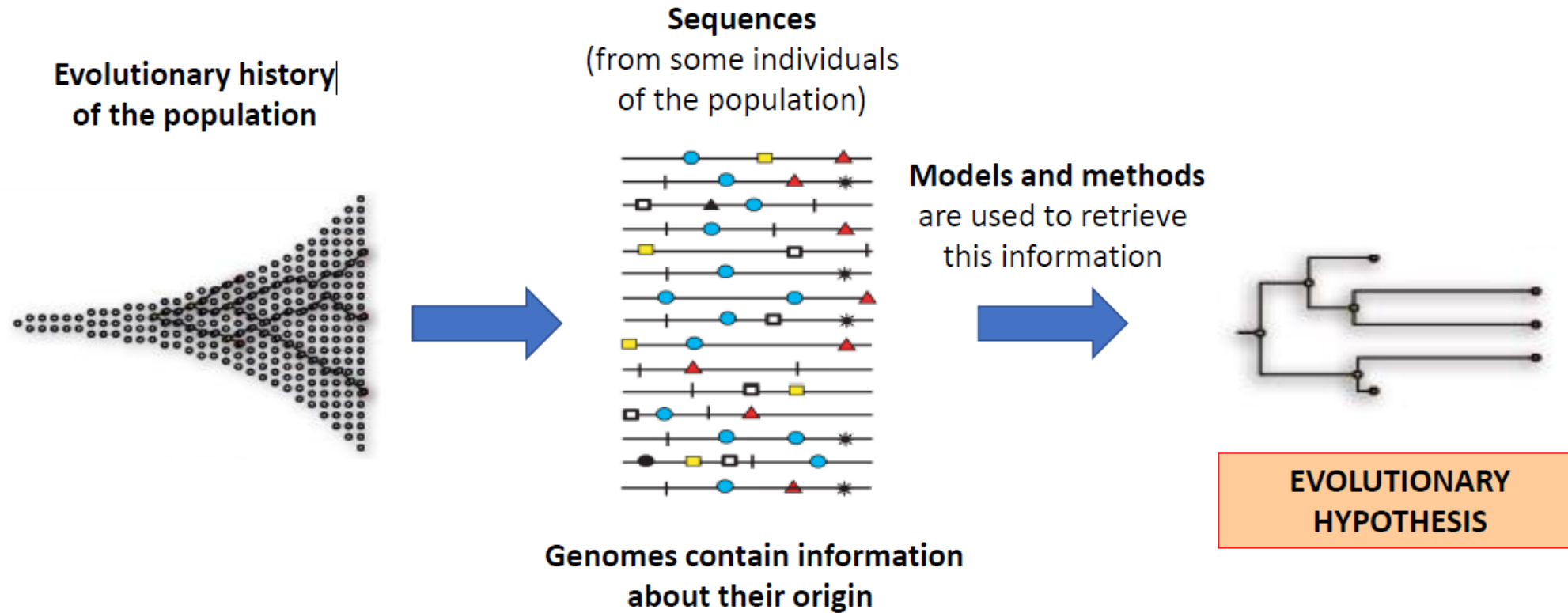


cgMLST and wgMLST



Core Genome + Accessory Genome = PanGenome -> wgMLST

Phylogenetic analysis



Phylogenetic analysis allows us to reconstruct the evolutionary history and study the processes that gave rise to it.

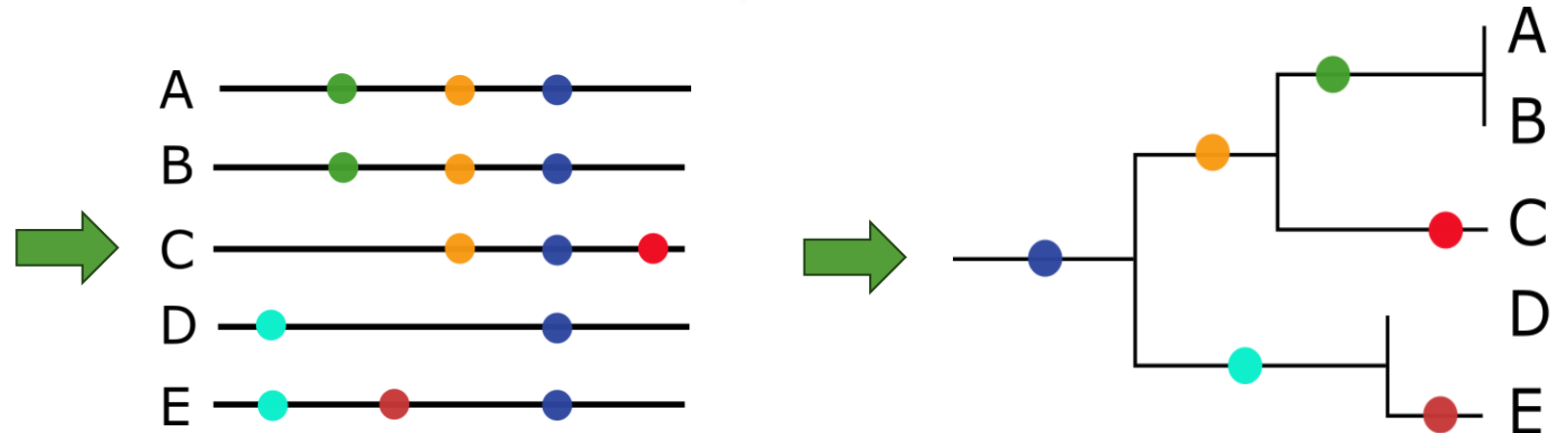
Análisis filogenético

Utiliza un **modelo matemático** para predecir cómo evolucionó probablemente la secuencia e inferir el árbol

3. Inferir relaciones evolutivas

Method	Data used	Tree search	Evolutionary Model
Distance	Pairwise distance	Simple algorithm	Can be complex
Parsimony *	All sites	Mainly hill climbing	Simple
Maximum likelihood *	All sites	Hill climbing	Can be complex
Bayesian inference *	All sites (+ other info)	MCMC	Can be very complex

* attempt to find the BEST tree



Historia evolutiva de la población

Secuencias

Hipotesis evolutiva

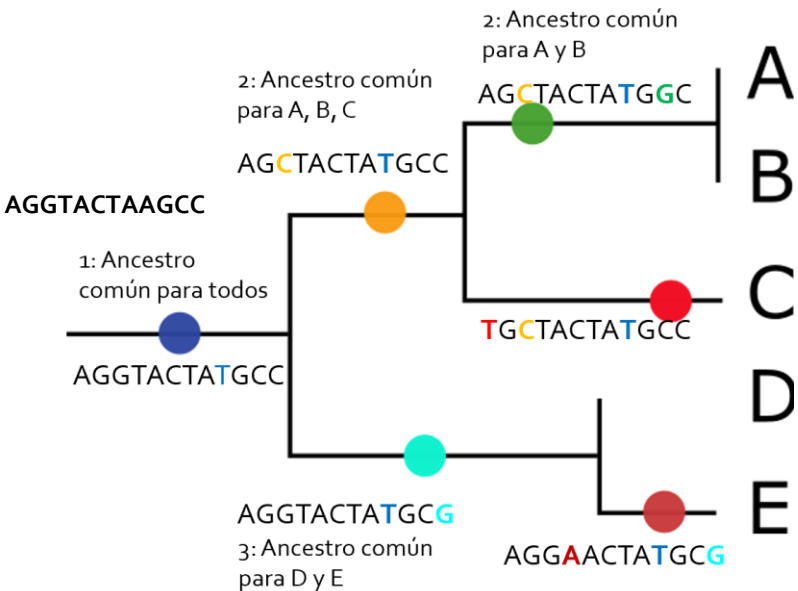
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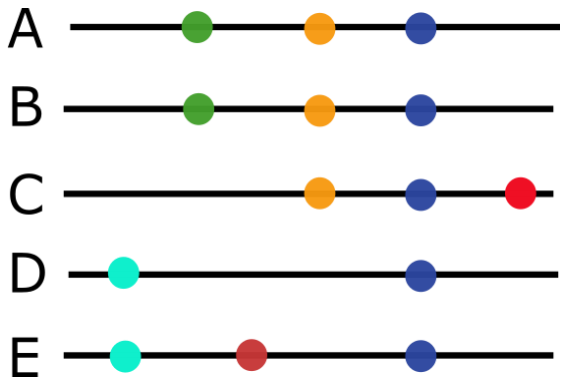
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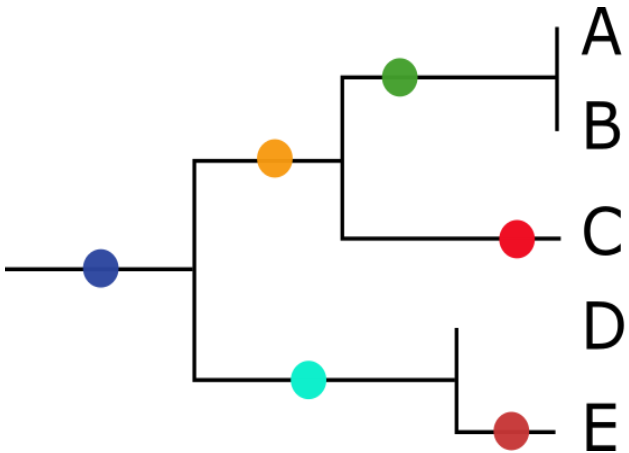
* attempt to find the BEST tree



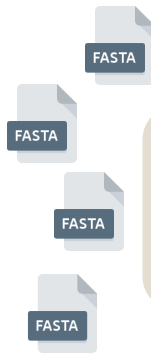
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Secuencias



Hipotesis evolutiva



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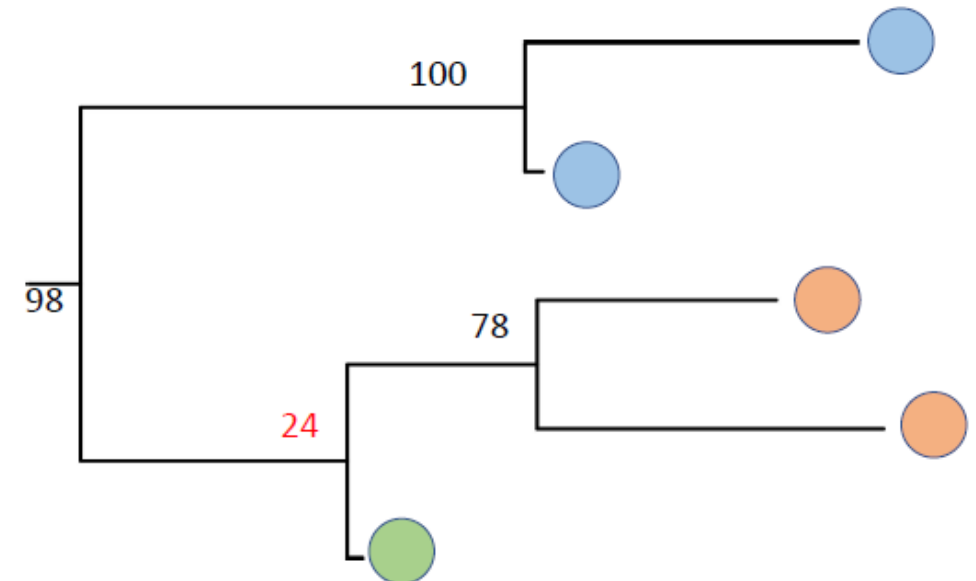
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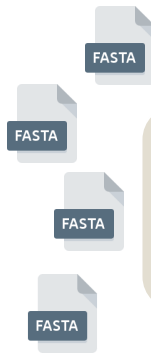
7. Visualize and export
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labels, colors, and
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Relate findings to the
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¿Qué tan robusto es el árbol?

- **Bootstrapping** es una técnica estadística que utiliza el **remuestreo aleatorio** de datos para determinar el error de muestreo.
- Da una idea de la **"fiabilidad"** de las ramas y los grupos.
- X número de bootstraps (réplicas remuestreadas) que se crean de sus datos de entrada (MSA)
- Normalmente ejecuta entre 100 y 1.000 bootstraps para el análisis **de ML**
- Normalmente se considera significativo cuando es superior al 70% (o 0.7 o 70/100)





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Herramientas para visualización de árboles filogenéticos

- FigTree(download, free): <http://tree.bio.ed.ac.uk/software/figtree/>
- Geneious(download, \$\$): <https://www.geneious.com/>
- UGENE (download, free): <http://ugene.net/>
- TreeView(download, free): <http://jtreeview.sourceforge.net/>
- iTOL(online, free or \$\$): <https://itol.embl.de/>
- ETE Toolkit (online, free): <http://etetoolkit.org/treeview/>
- MicroReact(online, free): <http://microreact.org/>

FASTA

FASTA

FASTA

FASTA

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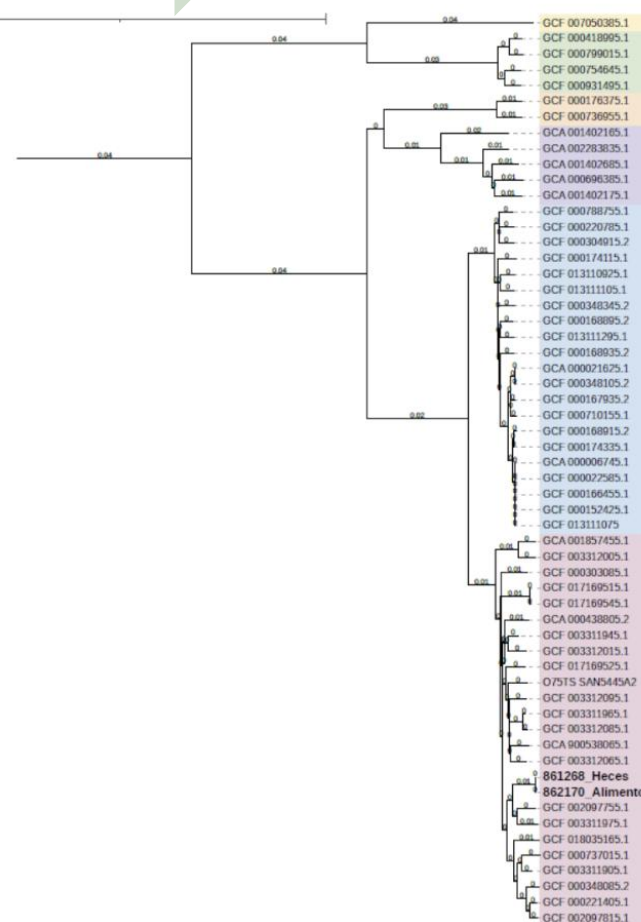
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Tree scale: 0.1

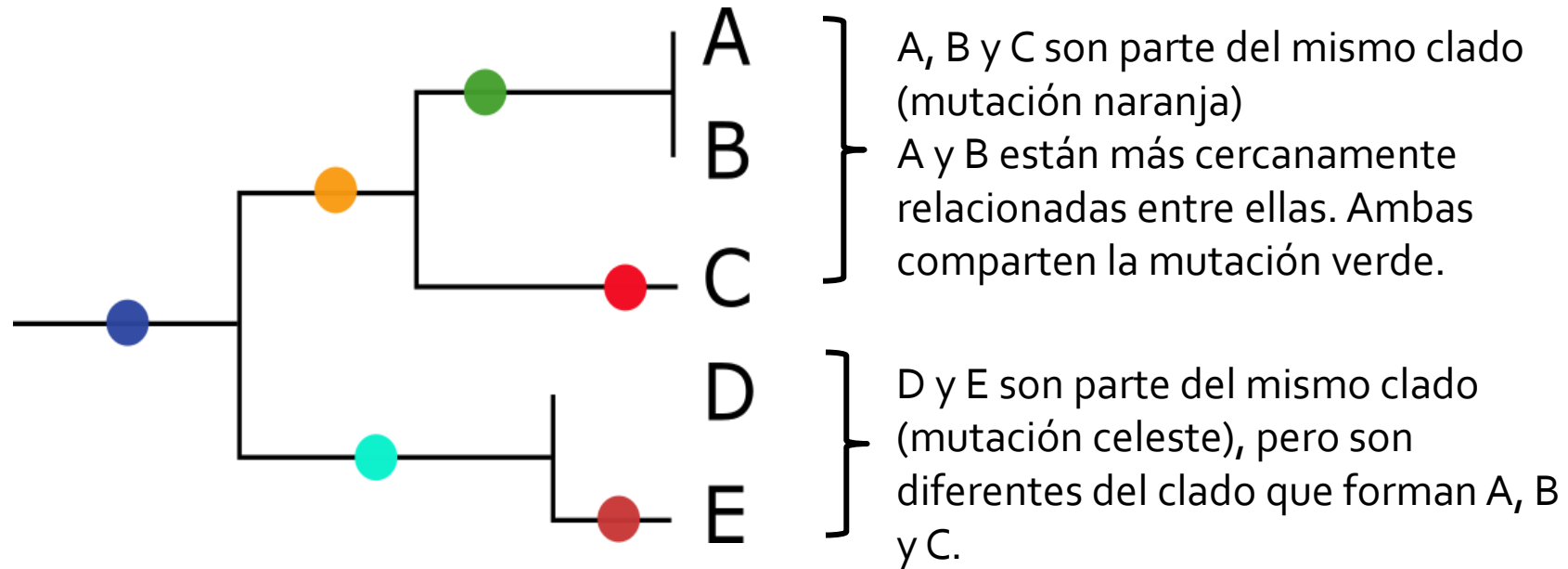


Species	
■	<i>Vibrio fluvialis</i>
■	<i>Vibrio furnissii</i>
■	<i>Vibrio mimicus</i>
■	<i>Vibrio metoecus</i>
■	<i>Vibrio cholerae</i>
■	<i>Vibrio paracholerae</i>

Interpretación de árboles filogenéticos

"La filogenia se aproxima a la epidemiología"

–Lee Katz



Por ejemplo, podríamos plantear la hipótesis de que A y B son parte del mismo evento de transmisión. Pero no podemos distinguir si hubo transmisión directa entre A y B o si fueron infectados por el mismo individuo.

La información epidemiológica puede apoyar, rechazar o refinar el modelo de transmisión.

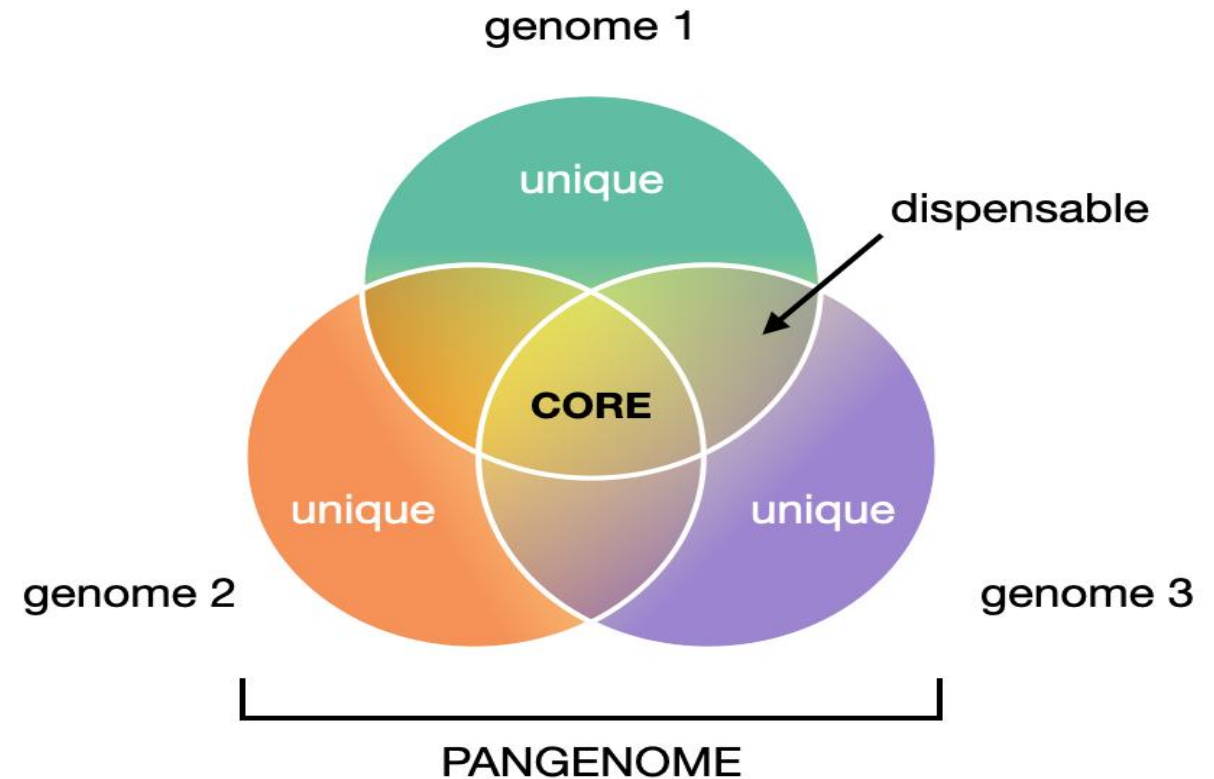
La filogenética puede ayudar a las investigaciones epidemiológicas relacionadas con los brotes

- Refinando el brote al descartar a los individuos dentro o fuera
- Generando hipótesis sobre la transmisión



Árbol SNP del Genoma Central vs Árbol SNP del Genoma Pan

- El genoma central de un conjunto de secuencias contiene sólo los **sitios compartidos por todas las secuencias**
 - Esto es muy útil al analizar un conjunto de secuencias donde se espera cierta diversidad (múltiples genotipos en el mismo árbol, o incluyendo/excluyendo muestras de un brote)
 - Es más rápido y menos intensivo desde el punto de vista computacional construir un árbol utilizando el genoma central
- El pangenoma contiene **todos los sitios** encontrados en todas las secuencias del conjunto
 - Es muy útil cuando se trata de evaluar la posible transmisión entre individuos, caracterizar un brote durante un largo periodo de tiempo o describir la estructura general de la población de un patógeno
 - El número de SNP en el pangenoma siempre será mayor que en el genoma central



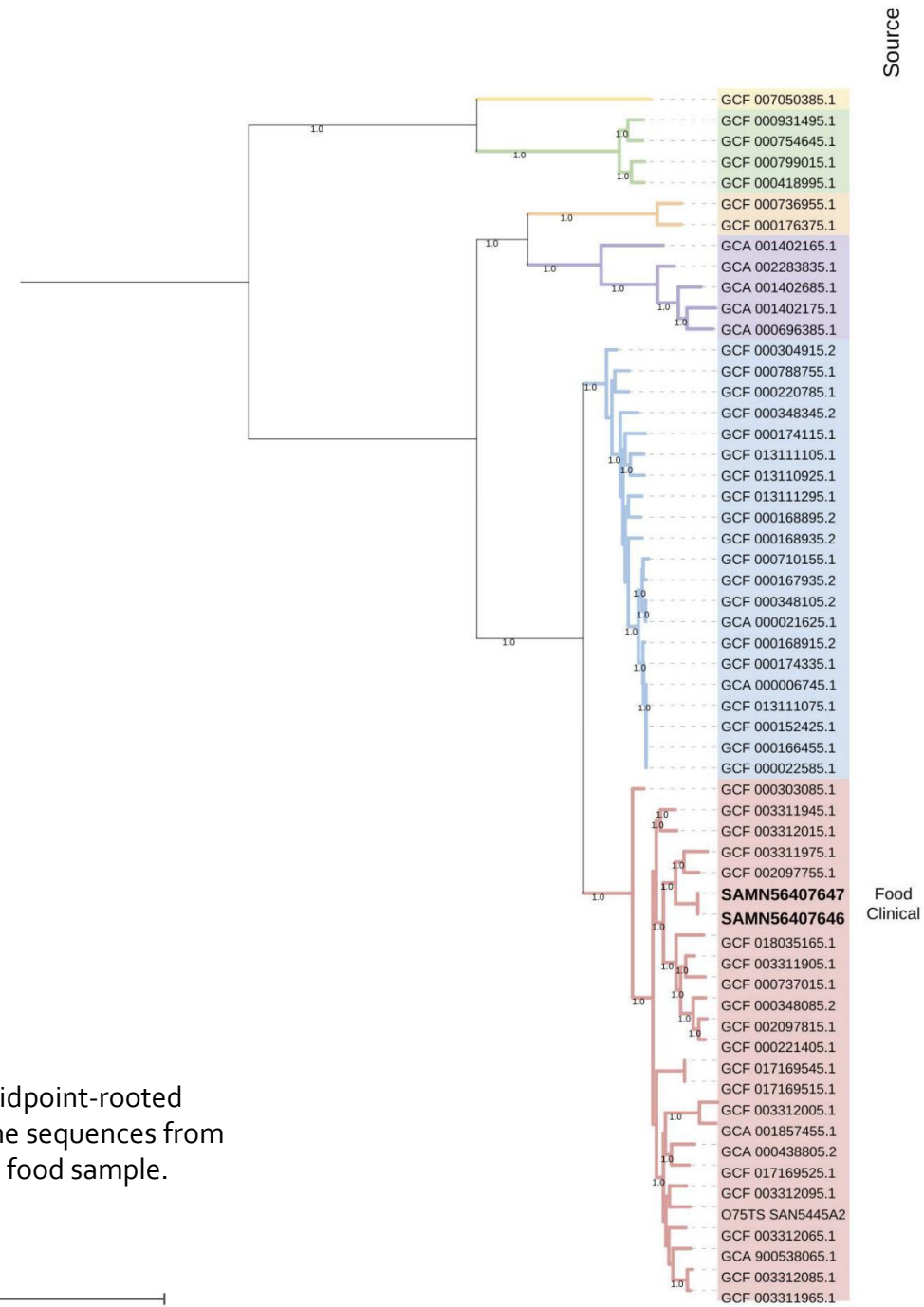
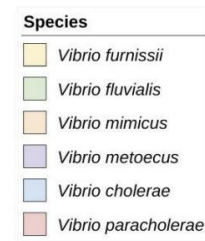


Figure 1. Neighbor-joining analysis for species identification of *Vibrio* sp. Midpoint-rooted phylogenetic tree showing the evolutionary relationships of 56 core genome sequences from different *Vibrio* species and 2 query sequences derived from a clinical and a food sample.

Tree scale: 0.1

